Lab 7 Plan

1. Understand ANSI escape codes and lookup tables, get the board to show up
2. Set up state machine with states: play, pause, powerup, game over
3. Set up global variables: lives, score, level, remaining pellets, positions (x and y) of Pacman and ghosts, direction
4. Get Pacman to move on the board
   - Pacman is character '<'
   - Using the UART interrupt, detect what keys have been pressed and that changes the direction variable
   - Using the timer interrupt based on the direction variable make Pacman move up, down, left, or right
   - If Pacman encounters a wall, it will not move until the direction variable is changed to a valid direction
5. Implement pause logic
   - On the SW1 interrupt, pause the game and update the state
   - Disable the timer interrupt when pausing, enable when unpausing
   - Ignore all UART interrupts when paused
6. Get Pacman to collect the pellets and increment the score
   - Pellets are '.' and every time Pacman moves to a spot with one, the score is incremented by 10 and the '.' is removed.
7. Get the ghosts to move, set up randomizer
   - Ghosts are character 'A' with different colors for each ghost
   - Ghosts move in a random valid direction when at an intersection
8. Losing life logic/ game over logic
   - Player is given 4 lives indicated by the LEDs on the daughter board
   - After losing a life, Pacman is moved back to the start, and an LED is turned off
   - Once all 4 lives are gone, the program ends and it is reset to the initial board
9. Set up powerup pellets, once activated make ghosts eatable
   - Power-up pellets are character 'O' and every time it is encountered, ghosts are made eatable and are changed to character 'W', every time a ghost is encountered under the influence, score is incremented by 100 and doubles for each ghost
   - LED is used on the Tiva to indicate the time the powerup lasts. Once it's close to ending, it will blink indicating that the timer is coming to an end.
10. Level complete logic
    - Move Pacman back to the start and reset the board to its original state.
    - Update the timer to interrupt 0.01 seconds sooner.